

Chaitanya Chaudhari

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EDUCATION

George Mason University

Master of Science in Computer Science

Fairfax, VA

Aug 2024 - May 2026

Savitribai Phule Pune University

Bachelor of Engineering in Computer Engineering

Pune, India

May 2020 - Jun 2024

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, Go, C++

AI/ML: Large Language Models (LLMs), RAG Pipelines, Vector Embeddings, Model Evaluation, Agentic AI, Prompt Engineering, Azure AI Foundry, Semantic Search

Backend & Systems: REST APIs, API Infrastructure, Distributed Systems, Data Pipelines, Systems Engineering, Microservices

Cloud & Infrastructure: Azure Pipelines, AWS (Lambda, EC2, S3, RDS, CloudWatch), Docker, Kubernetes, CI/CD

Databases: PostgreSQL, MySQL, MongoDB, Vector Databases, Amazon RDS

Certifications: Introduction to Azure DevOps, AWS ML and Cloud Foundations, OCI 2025 AI Foundations Associate

EXPERIENCE

Open Source Contributor | Secure Coding Case Studies

Feb 2026 – Apr 2026

- Researched CVE-2025-54416 (CVSS 9.1) in tj-actions/branch-names, identifying how eval re-parsed sanitized shell input to enable arbitrary code execution on CI/CD runners via malicious Git branch names.
- Published defense patterns and integration guidance covering environment variable isolation, SHA pinning, and least-privilege design, contributing research to a public secure engineering education repository.

Software Engineer | Rotaract Club of Nashik Nine Hills

Jul 2023 – Mar 2024

- Designed 12+ RESTful API endpoints and data pipelines within a microservices architecture serving 500+ users, maintaining consistent data flow and reliability across 4+ interconnected backend services
- Cut release-time configuration errors by 50% by containerizing backend services with Docker and integrating automated tests and data pipeline validation into CI/CD pipelines, ensuring reliable code integration across environments
- Tested end-to-end application workflows through systematic black-box and A/B testing, identified and resolved 20+ failures across API interactions and data consistency before production rollout

DevOps Intern | Cognifront Software Pvt. Ltd

Jan 2023 – Mar 2023

- Reduced manual build and test effort by 40% by automating CI/CD workflows and data pipeline validation using GitHub Actions across staging and production environments.
- Diagnosed and resolved 10+ production incidents through Linux log analysis and systematic root cause analysis, improving system uptime and reliability.
- Reduced infrastructure setup time by 50% by provisioning and documenting AWS infrastructure including EC2, S3, and IAM using Terraform across 3+ environments, creating reproducible, version-controlled configurations.

PROJECTS

AI Reasoning Agent for CI/CD | [GitHub](#) | Microsoft Agents League Hackathon

Jun 2026 – Jun 2026

- Engineered a multi-step reasoning agent on Azure AI Foundry using an o4-mini reasoning model, ingesting GitHub Actions failure logs and reasoning through a 4-stage chain of error classification, root cause analysis, and fix generation.
- Developed an evaluation layer validating agent outputs across 5+ real failure cases, achieving consistent structured fix suggestions and prototyping a research-driven approach to automated CI/CD incident resolution.

GMU SmartPatriot | [Live](#) | [GitHub](#) | PatriotHacks Hackathon

Mar 2026 - Jun 2026

- Constructed a conversational AI assistant using a RAG pipeline integrating LLM-based generation with Amazon Bedrock and vector search, improving response accuracy by 25% across 500+ queries through prompt engineering and retrieval optimization.
- Established an automated evaluation framework measuring precision, recall, and failure cases across 500+ test queries, reducing hallucination rates by 30% through systematic iteration and research-driven refinement.

MedLedger AI | [Live](#) | [GitHub](#) | HackFax Hackathon

Feb 2026 – Jun 2026

- Architected an LLM-powered data pipeline converting unstructured documents into vector embeddings, enabling contextual retrieval across 1,000+ records at sub-second latency with confidence scoring for reliability.
- Created API endpoints and a conversational AI interface combining AI-generated insights with structured database content, delivering accurate natural language queries over historical data.

Serverless ML Inference API | [GitHub](#)

Dec 2025 - Feb 2026

- Processed 5K+ inference requests at sub-3s latency by building an event-driven distributed AI pipeline on AWS Lambda with idempotent request handling and fault-tolerant design.
- Reduced issue diagnosis time by 40% by building reporting and monitoring mechanisms using CloudWatch logs, metrics, and automated alerting across distributed components.